



Database Management System Project

Check Point 4

[Topic : Queries]

GUJARAT TUORISM MANAGEMENT SYSTEM

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Q. List all tourist spots sorted by highest rating.

Query :

```
SELECT spot_name, category, rating, total_reviews
FROM TOURIST_SPOT
ORDER BY rating DESC;
```

Output:

	spot_name character varying (200)	category character varying (100)	rating numeric (3,1)	total_reviews integer
1	Somnath Temple	Religious	4.9	20000
2	Statue of Unity	Heritage	4.9	25000
3	Rann of Kutch	Natural	4.8	15000
4	Akshardham Temple	Religious	4.8	13000
5	Rani ki Vav	Heritage	4.8	7500
6	Gir National Park	Wildlife	4.7	11000
7	Sabarmati Ashram	Heritage	4.7	9800
8	Champaner-Pavagadh	Heritage	4.6	6000
9	Laxmi Vilas Palace	Heritage	4.6	5000
10	Modhera Sun Temple	Heritage	4.5	4000
11	Marine National Park	Natural	4.3	3000

Q. Find the 3 most visited tourist spots by total reviews.

Query:

```
SELECT spot_name, category, total_reviews
FROM TOURIST_SPOT
ORDER BY total_reviews DESC LIMIT 3;
```

Output:

	spot_name character varying (200)	category character varying (100)	total_reviews integer
1	Statue of Unity	Heritage	25000
2	Somnath Temple	Religious	20000
3	Rann of Kutch	Natural	15000

Q. Show all foreign tourists with their nationalities and contacts.

Query:

```
SELECT first_name, last_name, nationality, phone, email
FROM TOURIST WHERE tourist_type = 'FOREIGN'
ORDER BY nationality;
```

Output:

	first_name character varying (100)	last_name character varying (100)	nationality character varying (100)	phone character varying (20)	email character varying (200)
1	John	Smith	American	9876543214	john@gmail.com
2	Emma	Watson	British	9876543215	emma@gmail.com
3	Liu	Wei	Chinese	9876543217	liu@gmail.com
4	Carlos	Ruiz	Spanish	9876543219	carlos@gmail.com

Q. Find all UNESCO Heritage tourist spot.

Query:

```
SELECT spot_name, district_id, established_year, rating
FROM TOURIST_SPOT
WHERE is_unesco_heritage = TRUE;
```

Output:

	spot_name character varying (200)	district_id integer	established_year integer	rating numeric (3,1)
1	Rani ki Vav	7	1063	4.8
2	Champaner-Pavagadh	2	8	4.6

Q. List accommodation options with available rooms, sorted by price.

Query:

```
SELECT name, type, available_rooms, price_local, price_domestic,
price_foreign
FROM ACCOMMODATION
WHERE available_rooms > 0
ORDER BY price_local ASC;
```

Output:

	name character varying (200)	type character varying (100)	available_rooms integer	price_local double precision	price_domestic double precision	price_foreign double precision
1	Somnath Guest House	Guest House	10	800	1500	4000
2	Gandhinagar Tulip Inn	Hotel	20	1500	2500	6000
3	The Gateway Vadodara	Hotel	15	2500	4500	9000
4	Taj Vivanta Ahmedabad	Hotel	30	3000	5000	10000
5	Gir Jungle Lodge	Resort	8	3500	6000	13000
6	Kevadia Jungle Resort	Resort	12	3500	7000	15000
7	Rann Riders Resort	Resort	5	4000	6000	14000

Q. Total revenue collected from all visits.

Query:

```
SELECT SUM(total_fee_paid) AS total_visit_revenue  
FROM VISIT;
```

Output:

Data Output		Messages
	total_visit_revenue	double precision
1	8780	

Q. Count visits by tourist type.

Query:

```
SELECT tourist_type, COUNT(*) AS visit_count  
FROM VISIT  
GROUP BY tourist_type;
```

Output:

Data Output		Messages	Notifications
	tourist_type	visit_count	
	character varying (50)	bigint	
1	LOCAL	4	
2	FOREIGN	6	
3	DOMESTIC	5	

Q. Which tourist spot have average feedback rating above 4.

Query:

```
SELECT spot_id, AVG(rating) AS avg_feedback  
FROM FEEDBACK F  
JOIN VISIT V ON F.visit_id = V.visit_id  
GROUP BY V.spot_id HAVING AVG(rating) > 4;
```

Output:

Data Output		Messages	Notification
	spot_id	avg_feedback	
	integer	numeric	
1	10	5.0000000000000000	
2	11	4.3333333333333333	
3	7	5.0000000000000000	
4	3	5.0000000000000000	
5	1	5.0000000000000000	

Q. Average entry fee per tourist spot category

Query:

```
SELECT category,  
       AVG(entry_fee_local_inr) AS avg_local,  
       AVG(entry_fee_domestic_inr) AS avg_domestic,  
       AVG(entry_fee_foreign_inr) AS avg_foreign  
FROM TOURIST_SPOT  
GROUP BY category;
```

Output:

Data Output Messages Notifications				
Showing rows: 1 to 4				
	category character varying (100)	avg_local numeric	avg_domestic numeric	avg_foreign numeric
1	Wildlife	75.0000000000000000	500.0000000000000000	2000.0000000000000000
2	Heritage	71.6666666666666667	127.5000000000000000	500.0000000000000000
3	Religious	0.0000000000000000	0.0000000000000000	0.0000000000000000
4	Natural	75.0000000000000000	350.0000000000000000	1150.0000000000000000

Q. Find open complaints with more than 1 complaint per type.

Query:

```
SELECT complain_against, COUNT(*) AS complaint_count  
FROM FEEDBACK  
WHERE status = 'Open' OR status = 'Resolved'  
AND feedback_type = 'Complaint'  
GROUP BY complain_against  
HAVING COUNT(*) >= 1  
ORDER BY complaint_count DESC;
```

Output:

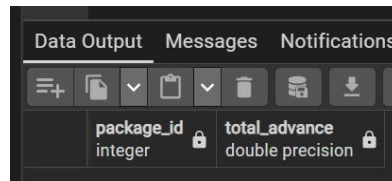
Data Output Messages Notifications		
	complain_against character varying (200)	complaint_count bigint
1	Management	2
2	Contractor	1
3	Guide	1
4	Staff	1
5	[null]	1

Q. Which tourist package have total bookings exceeding Rs. 20,000.

Query:

```
SELECT package_id, SUM(advance_payment) AS total_advance  
FROM PACKAGE_BOOKING  
GROUP BY package_id  
HAVING SUM(advance_payment) > 20000;
```

Output:



Data Output	Messages	Notifications
      		
package_id integer		total_advance double precision

Q. Show every tourist spot with its district name and region

Query:

```
SELECT S.spot_name, S.category, D.name AS district, D.region
FROM TOURIST_SPOT S
JOIN DISTRICT D ON S.district_id = D.district_id;
```

Output:

Data Output

Messages

Notifications

Showing rows: 1 to 11

	spot_name character varying (200)	category character varying (100)	district character varying (100)	region character varying (100)
1	Modhera Sun Temple	Heritage	Ahmedabad	Central Gujarat
2	Sabarmati Ashram	Heritage	Ahmedabad	Central Gujarat
3	Champaner-Pavagadh	Heritage	Vadodara	Central Gujarat
4	Laxmi Vilas Palace	Heritage	Vadodara	Central Gujarat
5	Rann of Kutch	Natural	Kutch	Saurashtra-Kutch
6	Somnath Temple	Religious	Somnath	Saurashtra-Kutch
7	Akshardham Temple	Religious	Gandhinagar	Central Gujarat
8	Marine National Park	Natural	Rajkot	Saurashtra-Kutch
9	Rani ki Vav	Heritage	Patan	North Gujarat
10	Gir National Park	Wildlife	Junagadh	Saurashtra-Kutch
11	Statue of Unity	Heritage	Narmada	South Gujarat

Q. List all confirmed package booking with package name.

Query:

```
SELECT PB.booking_id, TP.package_name, PB.booking_date,
       PB.group_size, PB.booking_status
FROM PACKAGE_BOOKING PB
JOIN TOURIST_PACKAGE TP ON PB.package_id = TP.package_id
WHERE PB.booking_status = 'Confirmed';
```

Output:

	booking_id integer	package_name character varying (200)	booking_date date	group_size integer	booking_status character varying (50)
1	1	Gujarat Heritage Circuit	2023-12-01	2	Confirmed
2	2	Rann Adventure Package	2024-10-15	1	Confirmed
3	3	Spiritual Gujarat Tour	2023-12-20	4	Confirmed
4	4	Wildlife & Nature Escape	2024-09-01	2	Confirmed
5	5	Gujarat Heritage Circuit	2023-12-10	3	Confirmed
6	7	Rann Adventure Package	2024-10-20	1	Confirmed
7	8	Spiritual Gujarat Tour	2024-01-05	1	Confirmed
8	10	Royal Vadodara Weekend	2024-03-10	3	Confirmed
9	11	Statue & Narmada Explo...	2024-04-20	1	Confirmed
10	12	Statue & Narmada Explo...	2024-05-01	2	Confirmed

Q. Show all guide assignments with guide name and spot name.

Query:

```
SELECT G.name AS guide_name, TS.spot_name,
       GA.assigned_date, GA.fee_charged
FROM GUIDE_ASSIGNMENT GA
JOIN GUIDE G ON GA.guide_id = G.guide_id
JOIN TOURIST_SPOT TS ON GA.spot_id = TS.spot_id;
```

Output:

	guide_name character varying (200)	spot_name character varying (200)	assigned_date date	fee_charged double precision
1	Harshad Mehta	Sabarmati Ashram	2024-01-10	600
2	Anjali Bhatt	Rann of Kutch	2024-01-12	2000
3	Seema Desai	Rani ki Vav	2024-02-05	500
4	Harshad Mehta	Gir National Park	2024-02-18	1500
5	Anjali Bhatt	Gir National Park	2024-03-15	2000
6	Zainab Sheikh	Rani ki Vav	2024-03-01	1800
7	Ravi Trivedi	Marine National Park	2024-04-10	300
8	Seema Desai	Champaner-Pavagadh	2024-05-01	500
9	Anjali Bhatt	Laxmi Vilas Palace	2024-02-10	800
10	Zainab Sheikh	Akshardham Temple	2024-03-10	1800
11	Harshad Mehta	Statue of Unity	2024-05-15	600
12	Anjali Bhatt	Statue of Unity	2024-06-10	800
13	Zainab Sheikh	Statue of Unity	2024-07-20	1800

Q. Show accommodation allotment with hotel name and package name.

Query:

```
SELECT A.name AS hotel, TP.package_name,
       AA.check_in_date, AA.check_out_date
FROM ACCOMMODATION_ALLOTMENT AA
JOIN ACCOMMODATION A ON AA.accommodation_id =
A.accommodation_id
JOIN TOURIST_PACKAGE TP ON AA.package_id = TP.package_id;
```

Output:

	hotel	package_name	check_in_date	check_out_date
	character varying (200)	character varying (200)	date	date
1	Taj Vivanta Ahmedabad	Gujarat Heritage Circuit	2024-01-10	2024-01-15
2	Rann Riders Resort	Rann Adventure Package	2024-11-05	2024-11-08
3	Gir Jungle Lodge	Wildlife & Nature Escape	2024-10-10	2024-10-14
4	The Gateway Vadodara	Royal Vadodara Weekend	2024-04-15	2024-04-17
5	Gandhinagar Tulip Inn	Gujarat Heritage Circuit	2024-02-01	2024-02-06
6	Somnath Guest House	Spiritual Gujarat Tour	2024-01-15	2024-01-19
7	Kevadia Jungle Resort	Statue & Narmada Explo...	2024-05-15	2024-05-18

Q. Full visit receipt: tourist name, spot, district, feedback rating.

Query:

```
SELECT T.first_name, T.last_name, TS.spot_name,
       D.name AS district, V.visit_date,
       V.total_fee_paid, FB.rating
FROM VISIT V
JOIN TOURIST T ON V.tourist_id = T.tourist_id
JOIN TOURIST_SPOT TS ON V.spot_id = TS.spot_id
JOIN DISTRICT D ON TS.district_id = D.district_id
LEFT JOIN FEEDBACK FB ON FB.visit_id = V.visit_id;
```

Output:

	first_name	last_name	spot_name	district	visit_date	total_fee_paid	rating
	character varying (100)	character varying (100)	character varying (200)	character varying (100)	date	double precision	integer
1	Ramesh	Patel	Sabarmati Ashram	Ahmedabad	2024-01-10	0	5
2	John	Smith	Rann of Kutch	Kutch	2024-01-12	1500	3
3	Amit	Sharma	Somnath Temple	Somnath	2024-01-15	0	5
4	Sunita	Verma	Rani ki Vav	Patan	2024-02-05	80	2
5	Emma	Watson	Laxmi Vilas Palace	Vadodara	2024-02-10	500	4
6	Kiran	Modi	Gir National Park	Junagadh	2024-02-18	400	1
7	Liu	Wei	Rani ki Vav	Patan	2024-03-01	1200	4
8	Meera	Joshi	Akshardham Temple	Gandhinagar	2024-03-10	0	5
9	Carlos	Ruiz	Rann of Kutch	Kutch	2024-03-22	1800	5
10	Priya	Shah	Modhera Sun Temple	Ahmedabad	2024-04-05	50	3
11	Emma	Watson	Marine National Park	Rajkot	2024-04-10	800	4
12	Amit	Sharma	Champaner-Pavagadh	Vadodara	2024-05-01	100	5
13	Ramesh	Patel	Statue of Unity	Narmada	2024-05-15	300	5
14	Sunita	Verma	Statue of Unity	Narmada	2024-06-10	1050	3
15	John	Smith	Statue of Unity	Narmada	2024-07-20	1000	5

Q. Which spot are included in each package, with district info?

Query:

```
SELECT TP.package_name, TS.spot_name, D.name AS district,
       PS.day_number, PS.visit_order_no
FROM PACKAGE_SPOT PS
JOIN TOURIST_PACKAGE TP ON PS.package_id = TP.package_id
```

```
JOIN TOURIST_SPOT TS ON PS.spot_id = TS.spot_id
JOIN DISTRICT D ON TS.district_id = D.district_id
ORDER BY TP.package_name, PS.day_number;
```

Output:

Data Output Messages Notifications					
	package_name character varying (200)	spot_name character varying (200)	district character varying (100)	day_number integer	visit_order_no integer
1	Gujarat Heritage Circuit	Sabarmati Ashram	Ahmedabad	1	1
2	Gujarat Heritage Circuit	Modhera Sun Temple	Ahmedabad	2	1
3	Gujarat Heritage Circuit	Rani ki Vav	Patan	3	1
4	Gujarat Heritage Circuit	Champaner-Pavagadh	Vadodara	4	1
5	Gujarat Heritage Circuit	Laxmi Vilas Palace	Vadodara	5	1
6	Rann Adventure Package	Rann of Kutch	Kutch	1	1
7	Rann Adventure Package	Marine National Park	Rajkot	3	2
8	Royal Vadodara Weekend	Laxmi Vilas Palace	Vadodara	1	1
9	Royal Vadodara Weekend	Champaner-Pavagadh	Vadodara	2	2
10	Spiritual Gujarat Tour	Somnath Temple	Somnath	1	1
11	Spiritual Gujarat Tour	Akshardham Temple	Gandhinagar	3	2
12	Statue & Narmada Explo...	Statue of Unity	Narmada	1	1
13	Statue & Narmada Explo...	Akshardham Temple	Gandhinagar	2	1
14	Statue & Narmada Explo...	Sabarmati Ashram	Ahmedabad	3	1
15	Wildlife & Nature Escape	Gir National Park	Junagadh	1	1
16	Wildlife & Nature Escape	Marine National Park	Rajkot	3	2

Q. Guide assignment report: guide name, spot, language spoken.

Query:

```
SELECT G.name, TS.spot_name, GA.assigned_date,
       GA.fee_charged, GL.language
FROM GUIDE_ASSIGNMENT GA
JOIN GUIDE G ON GA.guide_id = G.guide_id
JOIN TOURIST_SPOT TS ON GA.spot_id = TS.spot_id
JOIN GUIDE_LANGUAGE GL ON G.guide_id = GL.guide_id ORDER BY
G.name;
```

Output:

Data Output Messages Notifications						
	name character varying (200)	license_no character varying (100)	years_of_experience integer	spot_name character varying (200)	assigned_date date	fee_charged double precision
1	Anjali Bhatt	LIC-GUJ-003	15	Gir National Park	2024-03-15	2000
2	Anjali Bhatt	LIC-GUJ-003	15	Statue of Unity	2024-06-10	800
3	Anjali Bhatt	LIC-GUJ-003	15	Rann of Kutch	2024-01-12	2000
4	Anjali Bhatt	LIC-GUJ-003	15	Laxmi Vilas Palace	2024-02-10	800
5	Harshad Mehta	LIC-GUJ-001	12	Gir National Park	2024-02-18	1500
6	Harshad Mehta	LIC-GUJ-001	12	Statue of Unity	2024-05-15	600
7	Harshad Mehta	LIC-GUJ-001	12	Sabarmati Ashram	2024-01-10	600
8	Seema Desai	LIC-GUJ-002	8	Champaner-Pavagadh	2024-05-01	500
9	Seema Desai	LIC-GUJ-002	8	Rani ki Vav	2024-02-05	500
10	Zainab Sheikh	LIC-GUJ-004	10	Akshardham Temple	2024-03-10	1800
11	Zainab Sheikh	LIC-GUJ-004	10	Rani ki Vav	2024-03-01	1800
12	Zainab Sheikh	LIC-GUJ-004	10	Statue of Unity	2024-07-20	1800

Q. Transport route report: source and destination district names

Query:

```
SELECT S.name AS source, DE.name AS destination,  
       TR.transport_mode, TR.distance_km, TR.avg_duration_hrs  
FROM TRANSPORT_ROUTE TR  
JOIN DISTRICT S ON TR.source_district_id = S.district_id  
JOIN DISTRICT DE ON TR.destination_district_id = DE.district_id  
ORDER BY TR.distance_km;
```

Output:

	source character varying (100)	destination character varying (100)	transport_mode character varying (50)	distance_km double precision	avg_duration_hrs double precision
1	Ahmedabad	Gandhinagar	Road	30	0.5
2	Vadodara	Narmada	Rail	85	1
3	Vadodara	Narmada	Road	90	1.5
4	Somnath	Junagadh	Road	90	2
5	Gandhinagar	Patan	Road	100	2
6	Junagadh	Rajkot	Road	100	2.5
7	Ahmedabad	Vadodara	Road	110	2
8	Ahmedabad	Vadodara	Rail	110	1.5
9	Ahmedabad	Patan	Road	120	2.5
10	Narmada	Gandhinagar	Road	185	3
11	Ahmedabad	Narmada	Road	200	3.5
12	Rajkot	Kutch	Road	290	5
13	Vadodara	Somnath	Road	290	5
14	Kutch	Somnath	Road	320	5.5
15	Ahmedabad	Kutch	Road	400	7
16	Ahmedabad	Kutch	Air	400	1

Q. Event report: event name, host spot, and district.

Query:

```
SELECT E.event_name, E.start_date, E.end_date,  
       TS.spot_name, D.name AS district,  
       E.organizer_name, E.is_annual  
FROM EVENT E  
JOIN TOURIST_SPOT TS ON E.spot_id = TS.spot_id  
JOIN DISTRICT D ON TS.district_id = D.district_id ORDER BY E.start_date;
```

Output:

	event_name character varying (200)	start_date date	end_date date	spot_name character varying (200)	district character varying (100)	organizer_name character varying (200)	is_annual boolean
1	Champaner Heritage Walk	2024-01-15	2024-01-15	Champaner-Pavagadh	Vadodara	ASI Gujarat	false
2	Modhera Dance Festival	2024-01-19	2024-01-21	Modhera Sun Temple	Ahmedabad	Dept. of Culture Gujarat	true
3	Gir Wildlife Photo Exhibition	2024-02-01	2024-02-28	Gir National Park	Junagadh	Wildlife Trust India	false
4	Marine Bio Awareness Camp	2024-06-01	2024-06-05	Marine National Park	Rajkot	Forest Dept. Gujarat	false
5	Navratri Festival	2024-10-03	2024-10-12	Sabarmati Ashram	Ahmedabad	Ahmedabad Municipal	true
6	Sardar Patel Jayanti Celebration	2024-10-31	2024-10-31	Statue of Unity	Narmada	Govt. of Gujarat	true
7	Rann Utsav	2024-11-01	2025-02-28	Rann of Kutch	Kutch	Gujarat Tourism	true
8	Unity Marathon	2024-11-15	2024-11-15	Statue of Unity	Narmada	Kevadia Dev Authority	false
9	Laser & Light Show - Valley of Flow...	2024-12-01	2024-12-31	Statue of Unity	Narmada	SSNNL Gujarat	false

Q. Certified guides and the spots they have been assigned to.

Query:

```
SELECT G.name, G.license_no, G.years_of_experience,
       TS.spot_name, GA.assigned_date, GA.fee_charged
FROM GUIDE G
JOIN GUIDE_ASSIGNMENT GA ON G.guide_id = GA.guide_id
JOIN TOURIST_SPOT TS ON GA.spot_id = TS.spot_id
WHERE G.is_certified = TRUE
ORDER BY G.name;
```

Output:

	name character varying (200)	license_no character varying (100)	years_of_experience integer	spot_name character varying (200)	assigned_date date	fee_charged double precision
1	Anjali Bhatt	LIC-GUJ-003	15	Gir National Park	2024-03-15	2000
2	Anjali Bhatt	LIC-GUJ-003	15	Statue of Unity	2024-06-10	800
3	Anjali Bhatt	LIC-GUJ-003	15	Rann of Kutch	2024-01-12	2000
4	Anjali Bhatt	LIC-GUJ-003	15	Laxmi Vilas Palace	2024-02-10	800
5	Harshad Mehta	LIC-GUJ-001	12	Gir National Park	2024-02-18	1500
6	Harshad Mehta	LIC-GUJ-001	12	Statue of Unity	2024-05-15	600
7	Harshad Mehta	LIC-GUJ-001	12	Sabarmati Ashram	2024-01-10	600
8	Seema Desai	LIC-GUJ-002	8	Champaner-Pavagadh	2024-05-01	500
9	Seema Desai	LIC-GUJ-002	8	Rani ki Vav	2024-02-05	500
10	Zainab Sheikh	LIC-GUJ-004	10	Akshardham Temple	2024-03-10	1800
11	Zainab Sheikh	LIC-GUJ-004	10	Rani ki Vav	2024-03-01	1800
12	Zainab Sheikh	LIC-GUJ-004	10	Statue of Unity	2024-07-20	1800

Q. Find tourist who has never made a package booking

Query:

```
SELECT T.first_name, T.last_name, T.tourist_type, T.nationality
FROM TOURIST T
LEFT JOIN PACKAGE_BOOKING PB ON T.tourist_id = PB.tourist_id
WHERE PB.booking_id IS NULL;
```

Output:

Data Output

Messages

Notifications

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SQL

Showing rows: 1 to

	first_name character varying (100) 🔒	last_name character varying (100) 🔒	tourist_type character varying (50) 🔒	nationality character varying (100) 🔒
1	Deepak	Nair	DOMESTIC	Indian

Q. List all spots and any guide assignments, including unassigned spots.

Query:

```
SELECT TS.spot_name, TS.category,
       G.name AS guide_name, GA.assigned_date
FROM TOURIST_SPOT TS
LEFT JOIN GUIDE_ASSIGNMENT GA ON TS.spot_id = GA.spot_id
LEFT JOIN GUIDE G ON GA.guide_id = G.guide_id;
```

Output:

Data Output

Messages

Notifications

Q. Find tourist who visited a spot but left no feedback.

Query:

```
SELECT T.first_name, T.last_name, TS.spot_name, V.visit_date
FROM VISIT V
JOIN TOURIST T ON V.tourist_id = T.tourist_id
JOIN TOURIST_SPOT TS ON V.spot_id = TS.spot_id
LEFT JOIN FEEDBACK FB ON FB.visit_id = V.visit_id
WHERE FB.feedback_id IS NULL;
```

Output:

Data Output

Messages

Notifications

Q. List all guides and total assignments, including guides with zero assignment.

Query:

```
SELECT G.name, G.is_certified,
       COUNT(GA.assigned_date) AS total_assignments
FROM GUIDE G
LEFT JOIN GUIDE_ASSIGNMENT GA ON G.guide_id = GA.guide_id
GROUP BY G.guide_id, G.name, G.is_certified
ORDER BY total_assignments DESC;
```

Output:

	name	is_certified	total_assignments
	character varying (200)	boolean	bigint
1	Anjali Bhatt	true	4
2	Zainab Sheikh	true	3
3	Harshad Mehta	true	3
4	Seema Desai	true	2
5	Ravi Trivedi	false	1
6	Mohan Das	false	0

Q. Get a single list of all emails: tourists and guides.

Query:

```
SELECT email, 'Tourist' AS role FROM TOURIST WHERE email IS NOT
NULL
UNION
SELECT contact_no, 'Guide' AS role FROM GUIDE WHERE contact_no IS
NOT NULL;
```

Output:

	Data Output	Messages	Notifications
	email character varying	role text	
1	liu@gmail.com	Touri...	
2	ramesh@gmail.c...	Touri...	
3	8000100006	Guide	
4	kiran@gmail.com	Touri...	
5	8000100002	Guide	
6	8000100003	Guide	
7	meera@gmail.com	Touri...	
8	amit@gmail.com	Touri...	
9	deepak@gmail.co...	Touri...	
10	8000100001	Guide	
11	sunita@gmail.com	Touri...	
12	8000100004	Guide	
13	priya@gmail.com	Touri...	
14	emma@gmail.com	Touri...	
15	carlos@gmail.com	Touri...	
16	8000100005	Guide	
17	john@gmail.com	Touri...	

Q. Find the tourist spot with the highest average visit duration.

Query:

```
SELECT spot_name, category, avg_visit_duration
FROM TOURIST_SPOT
WHERE avg_visit_duration = (
    SELECT MAX(avg_visit_duration) FROM TOURIST_SPOT
);
```

Output:

	Data Output	Messages	Notifications
	spot_name character varying (200)	category character varying (100)	avg_visit_duration numeric (5,2)
1	Gir National Park	Wildlife	5.00

Q. Find guides whose fees for the foreign tourists exceeds the average.

Query:

```
SELECT name, is_certified, fees_foreign
FROM GUIDE
WHERE fees_foreign > (
    SELECT AVG(fees_foreign) FROM GUIDE
)
ORDER BY fees_foreign DESC;
```

Output:

	name character varying (200)	is_certified boolean	fees_foreign double precision
1	Anjali Bhatt	true	2000
2	Zainab Sheikh	true	1800
3	Harshad Mehta	true	1500

Q. Find packages where total amount to be paid exceeds average (correlated)

Query:

```
SELECT PB.booking_id, PB.package_id,
       PB.tourist_id, PB.amount_to_be_paid
FROM PACKAGE_BOOKING PB
WHERE PB.amount_to_be_paid > (
    SELECT AVG(PB2.amount_to_be_paid)
    FROM PACKAGE_BOOKING PB2
    WHERE PB2.package_id = PB.package_id
);
```

Output:

	booking_id [PK] integer	package_id integer	tourist_id integer	amount_to_be_paid double precision
1	3	3	4	12000
2	4	4	8	36000
3	5	1	6	24000
4	6	5	9	20000
5	12	6	8	26000

Q. Find guide who speaks all languages in the system.

Query:

```
SELECT G.name
FROM GUIDE G
WHERE NOT EXISTS (
    SELECT GL_all.language
    FROM GUIDE_LANGUAGE GL_all
    WHERE NOT EXISTS (
        SELECT 1
        FROM GUIDE_LANGUAGE GL
        WHERE GL.guide_id = G.guide_id
        AND GL.language = GL_all.language
    )
);
```

Output:

Data Output		Messages	Notifications
	name character varying (200)		
1	Zainab Sheikh		

Q. Find tourists who have visited all natural spots.

Query:

```
SELECT T.first_name, T.last_name
FROM TOURIST T
WHERE NOT EXISTS (
    SELECT TS.spot_id
    FROM TOURIST_SPOT TS
    WHERE TS.category = 'Natural'
    AND NOT EXISTS (
        SELECT 1
        FROM VISIT V
        WHERE V.tourist_id = T.tourist_id
        AND V.spot_id = TS.spot_id
    )
);
```

Output:

first_name		last_name	
character varying (100)		character varying (100)	

Q. Rank tourist spots by total revenue collected from visits.

Query:

```
SELECT TS.spot_name, TS.category,
       SUM(V.total_fee_paid) AS total_revenue,
       COUNT(V.visit_id) AS visit_count
FROM TOURIST_SPOT TS
JOIN VISIT V ON TS.spot_id = V.spot_id
GROUP BY TS.spot_id, TS.spot_name, TS.category
ORDER BY total_revenue DESC;
```

Output:

Data Output Messages Notifications				
	spot_name character varying (200)	category character varying (100)	total_revenue double precision	visit_count bigint
1	Rann of Kutch	Natural	3300	2
2	Statue of Unity	Heritage	2350	3
3	Rani ki Vav	Heritage	1280	2
4	Marine National Park	Natural	800	1
5	Laxmi Vilas Palace	Heritage	500	1
6	Gir National Park	Wildlife	400	1
7	Champaner-Pavagadh	Heritage	100	1
8	Modhera Sun Temple	Heritage	50	1
9	Sabarmati Ashram	Heritage	0	1
10	Somnath Temple	Religious	0	1
11	Akshardham Temple	Religious	0	1

Q. Tourist with confirm bookings but no visits in 2024.

Query:

```
SELECT T.first_name, T.last_name, T.tourist_type,
       PB.package_id, PB.tour_start_date
FROM TOURIST T
JOIN PACKAGE_BOOKING PB ON T.tourist_id = PB.tourist_id
WHERE PB.booking_status = 'Confirmed'
      AND T.tourist_id NOT IN (
        SELECT tourist_id
        FROM VISIT
        WHERE visit_date BETWEEN '2024-01-01' AND '2024-12-31'
      );
```

Output:

	first_name character varying (100)	last_name character varying (100)	tourist_type character varying (50)	package_id integer	tour_start_date date
--	---------------------------------------	--------------------------------------	--	-----------------------	-------------------------